

On multiple choice questions there is exact one right answer. Please check the right answer. You are permitted to check only one answer. If you check more than one correct answer you will get 0 points. If you check the right answer you will get maximal points. Exam note cross must be set correctly.

QUESTION 1 OF 20

Marked out of 3.00

Equivalence partitioning is a technique for ...

- ☒ test data selection.
- ☐ test execution schedule selection.
- ☐ test class selection.
- ☐ test tool selection.

QUESTION 2 OF 20

Marked out of 3.00

You are to write test cases for system testing. When can this activity begin?

- ☐ during the system test
- ☒ as soon as the component test is completed
- ☐ shortly before the system test
- ☐ as soon as the requirements and architecture specification make this possible

QUESTION 3 OF 20

Marked out of 3.00

Which aspects can static quality assurance techniques be applied to?

- ☐ only activities within the entire software development process
- ☐ only test cases, architecture plans, and milestone plans
- ☐ only artifacts within the entire software development process
- ☒ both artifacts and activities within the entire software development process

QUESTION 4 OF 20

Marked out of 3.00

Which of the measures listed is not part of the technical measures within constructive quality management?

- ☐ use of tools for code analysis
- ☐ use of modeling languages
- ☐ state-based testing
- ☒ choice of development environment

QUESTION 5 OF 20

Marked out of 3.00

There is a function in your software that frequently renders errors during the initial tests. What should you do in the next round of testing?

- ☐ Leave the function out of the test: it was sufficiently tested in the last round of testing.
- ☐ Test the function more extensively: this is only to verify the errors found and exclude the possibility that a mistake was made in the test execution.
- ☒ Test the function more extensively. On the basis of "error accumulation," more testing is required in the environment of the errors.
- ☐ Test the function to the same extent as before; after all, a sufficient number of errors have been found this way.

QUESTION 6 OF 20

Marked out of 3.00

The principle of early error detection and correction enables ...

- ☒ quality assurance of artifacts at the earliest possible time.
- ☐ quicker development.
- ☐ better planning of test resources and better use of infrastructure.
- ☐ improvement of the quality assurance position.

QUESTION 7 OF 20

Marked out of 3.00

In a conversation, your colleague states that component tests are a static measure.

What would be your objection?

Answer choice 1: Inquire whether the colleague might be confusing component tests with static code analysis, as static code analysis is a static measure.

Answer choice 2: Clarify that a component test is a dynamic measure because it involves executing the actual code, which is inherently dynamic and not static.

- ☒ only 2 is correct
- ☐ only 1 is correct
- ☐ neither 1 nor 2 is correct
- ☐ both 1 and 2 are correct

QUESTION 8 OF 20

Marked out of 3.00

In which software integration strategy are the selection criteria, such as component importance and relevance of individual software components to the project's goals, particularly important?

- ☐ big bang
- ☐ top-down
- ☐ bottom-up
- ☒ by-value

QUESTION 9 OF 20

Marked out of 3.00

When examining requirements, which activity do you not need to pay attention to?

- ☐ revision of the requirements
- ☐ organization of the required infrastructure
- ☐ documentation of the results
- ☒ coordination of external experts

QUESTION 10 OF 20

Marked out of 3.00

What is one advantage of independent quality assurance?

- ☐ The development team feels more responsible for quality when quality assurance is outsourced.
- ☐ The independence of the testers allows for outsourcing to another company with less communication overhead to clarify specifications.
- ☒ The independence of the testers promotes objectivity and brings another perspective on the specification during the evaluation.
- ☐ Developers will write more automated test cases to support the independent testers.

QUESTION 11 OF 20

Marked out of 3.00

Running through loops in a program code, sometimes several times, is particularly tested during ...

- ☐ acceptance tests.
- ☐ instruction tests.
- ☒ branch tests.
- ☐ condition tests.

QUESTION 12 OF 20

Marked out of 3.00

You use ISO 9126 as a quality model. Which area of quality management do you also have to consider with another quality model or a comparable one?

- ☐ None; ISO 9126 covers both product and process quality areas fully.
- ☐ the product-oriented quality area
- ☒ the process-oriented quality area
- ☐ Both product and process quality areas are not fully covered by ISO 9126.

QUESTION 13 OF 20

Marked out of 3.00

You would like to purchase a foosball table for your team, but management is not immediately interested. Which is your line of argument?

- ☐ A reward for good performance emphasizes employee value within the company.
- ☐ Creating a better working environment is an analytical measure; team spirit increases quality.
- ☐ Creating a better working environment is a constructive measure for quality improvement; quality is increased through team spirit.
- ☒ The purchase would foster informal collaboration among team members.

QUESTION 14 OF 20

Marked out of 3.00

Choose the best description of the difference between drivers and dummies (also known as stubs).

- ☐ Driver and dummy are synonyms.
- ☒ A driver accesses the function being tested, while a dummy is accessed by the function under test.
- ☐ A dummy accesses the function being tested, while a driver is accessed by the function being tested.
- ☐ A dummy simulates one component, while a driver represents multiple components.

QUESTION 15 OF 20

Marked out of 6.00

List two reasons why the complete testing of information systems is not possible.

1. Complete testing would mean trying all input combinations
and all possible system states.

2. There are too many test cases and it is not realistic to test
every user and situation.

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QUESTION 16 OF 20

Marked out of 6.00

Provide two examples of programming conventions.

1. *Naming of classes and methods should follow a fixed naming convention.*

2. *Use short variable names so the code is faster to read.*

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QUESTION 17 OF 20

Marked out of 8.00

- a) What is the difference between bytecode and source code analysis?
b) Using an example, explain why no automatic correction can be made for these two types of analysis.

a) Source code analysis checks the code written by the programmer. Bytecode analysis checks the compiled/intermediate code.

b) For example, an empty switch may look like a problem. A tool could remove it, but the empty block can be intentional, so automatic correction may change the meaning.

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QUESTION 18 OF 20

Marked out of 8.00

Explain the difference between templates and checklists, and demonstrate the use of both constructive measures.

A checklist is a list of things to check for a task, while a template gives a fixed format for a document.

A checklist helps make sure steps are not forgotten. A template keeps reports or specifications in the same structure.

Example: checklist for review steps; template for a status report. Both help to keep quality more consistent.

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QUESTION 19 OF 20

Marked out of 10.00

List and explain the two time-related options for quality assurance of architectures.

Ex ante means checking the architecture before implementation, to see if it fits the requirements and quality attributes.

Ex post is checking during or after implementation whether the result follows the architecture that was defined.

It can also show if users like the architecture, so it is useful after the system is finished.

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QUESTION 20 OF 20

Marked out of 10.00

You are reviewing a GUI where the overall layout is understandable, the color scheme of the interface is pleasing, and it meets the guidelines provided for delimiting foreground and background color. However, the individual input fields are not arranged in a series, and the necessary activities for name-checking are missing.

How would you formulate feedback? Use the feedback burger method.

Top bun: I like the overall layout and the colour scheme. The foreground and background contrast is also good.

Filling: The input fields are not in a clear sequence, so the user flow can be confusing. I would arrange them in a logical order.

The name-checking part is missing. I think this should be added so the entered names are correct.

Bottom bun: Overall the GUI is understandable and looks good. With these changes the usability should be better.

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This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run horizontally across the page. In the bottom right corner, there is a faint, diagonal watermark that reads "Distan".